Assignment 6

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Problem 1

```
data(prostate)
head(prostate)
```

```
##       lcavol lweight age      lbph svi      lcp gleason pgg45     lpsa
## 1 -0.5798185  2.7695  50 -1.386294   0 -1.38629       6     0 -0.43078
## 2 -0.9942523  3.3196  58 -1.386294   0 -1.38629       6     0 -0.16252
## 3 -0.5108256  2.6912  74 -1.386294   0 -1.38629       7    20 -0.16252
## 4 -1.2039728  3.2828  58 -1.386294   0 -1.38629       6     0 -0.16252
## 5  0.7514161  3.4324  62 -1.386294   0 -1.38629       6     0  0.37156
## 6 -1.0498221  3.2288  50 -1.386294   0 -1.38629       6     0  0.76547
```

```
str(prostate)
```

```
## 'data.frame':    97 obs. of  9 variables:
##  $ lcavol : num  -0.58 -0.994 -0.511 -1.204 0.751 ...
##  $ lweight: num  2.77 3.32 2.69 3.28 3.43 ...
##  $ age    : int  50 58 74 58 62 50 64 58 47 63 ...
##  $ lbph   : num  -1.39 -1.39 -1.39 -1.39 -1.39 ...
##  $ svi    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ lcp    : num  -1.39 -1.39 -1.39 -1.39 -1.39 ...
##  $ gleason: int  6 6 7 6 6 6 6 6 6 6 ...
##  $ pgg45  : int  0 0 20 0 0 0 0 0 0 0 ...
##  $ lpsa   : num  -0.431 -0.163 -0.163 -0.163 0.372 ...
```

```
f1 <- lm(lpsa ~ ., data=prostate)
summary(f1)
```

```
##
## Call:
## lm(formula = lpsa ~ ., data = prostate)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7331 -0.3713 -0.0170  0.4141  1.6381
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)  0.669337   1.296387   0.516       0.60693
## lcavol       0.587022   0.087920   6.677 0.00000000211 ***
## lweight      0.454467   0.170012   2.673       0.00896 **
## age         -0.019637   0.011173  -1.758       0.08229 .
## lbph         0.107054   0.058449   1.832       0.07040 .
```

```
## svi          0.766157   0.244309   3.136       0.00233 **
## lcp         -0.105474   0.091013  -1.159       0.24964
## gleason      0.045142   0.157465   0.287       0.77503
## pgg45        0.004525   0.004421   1.024       0.30886
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7084 on 88 degrees of freedom
## Multiple R-squared:  0.6548, Adjusted R-squared:  0.6234
## F-statistic: 20.86 on 8 and 88 DF,  p-value: < 2.2e-16
```

a-) Backward elimination

We will eliminate the largest p-value among those over 0.05, one at a time.

```
f11 <- update(f1, .~.-gleason)
summary(f11)
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol + lweight + age + lbph + svi + lcp +
##     pgg45, data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.73117 -0.38137 -0.01728  0.43364  1.63513
##
## Coefficients:
##              Estimate Std. Error t value       Pr(>|t|)
## (Intercept)  0.953926   0.829439   1.150        0.25319
## lcavol       0.591615   0.086001   6.879 0.000000000807 ***
## lweight      0.448292   0.167771   2.672        0.00897 **
## age         -0.019336   0.011066  -1.747        0.08402 .
## lbph         0.107671   0.058108   1.853        0.06720 .
## svi          0.757734   0.241282   3.140        0.00229 **
## lcp         -0.104482   0.090478  -1.155        0.25127
## pgg45        0.005318   0.003433   1.549        0.12488
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7048 on 89 degrees of freedom
## Multiple R-squared:  0.6544, Adjusted R-squared:  0.6273
## F-statistic: 24.08 on 7 and 89 DF,  p-value: < 2.2e-16
```

```
f12 <- update(f11, .~.-lcp)
summary(f12)
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol + lweight + age + lbph + svi + pgg45,
##     data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.77711 -0.41708  0.00002  0.40676  1.59681
##
```

```
## Coefficients:
##              Estimate Std. Error t value       Pr(>|t|)
## (Intercept)  0.980085   0.830665   1.180        0.24116
## lcavol       0.545770   0.076431   7.141 0.000000000231 ***
## lweight      0.449450   0.168078   2.674        0.00890 **
## age         -0.017470   0.010967  -1.593        0.11469
## lbph         0.105755   0.058191   1.817        0.07249 .
## svi          0.641666   0.219757   2.920        0.00442 **
## pgg45        0.003528   0.003068   1.150        0.25331
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7061 on 90 degrees of freedom
## Multiple R-squared:  0.6493, Adjusted R-squared:  0.6259
## F-statistic: 27.77 on 6 and 90 DF,  p-value: < 2.2e-16
```

```r
f13 <- update(f12, .~.-pgg45)
summary(f13)
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol + lweight + age + lbph + svi, data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.83505 -0.39396  0.00414  0.46336  1.57888
##
## Coefficients:
##             Estimate Std. Error t value        Pr(>|t|)
## (Intercept)  0.95100    0.83175   1.143        0.255882
## lcavol       0.56561    0.07459   7.583 0.0000000000277 ***
## lweight      0.42369    0.16687   2.539        0.012814 *
## age         -0.01489    0.01075  -1.385        0.169528
## lbph         0.11184    0.05805   1.927        0.057160 .
## svi          0.72095    0.20902   3.449        0.000854 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7073 on 91 degrees of freedom
## Multiple R-squared:  0.6441, Adjusted R-squared:  0.6245
## F-statistic: 32.94 on 5 and 91 DF,  p-value: < 2.2e-16
```

```r
f14 <- update(f13, .~.-age)
summary(f14)
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol + lweight + lbph + svi, data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.82653 -0.42270  0.04362  0.47041  1.48530
##
## Coefficients:
##             Estimate Std. Error t value        Pr(>|t|)
```

```
## (Intercept)  0.14554    0.59747   0.244         0.80809
## lcavol       0.54960    0.07406   7.422 0.0000000000564 ***
## lweight      0.39088    0.16600   2.355         0.02067 *
## lbph         0.09009    0.05617   1.604         0.11213
## svi          0.71174    0.20996   3.390         0.00103 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7108 on 92 degrees of freedom
## Multiple R-squared:  0.6366, Adjusted R-squared:  0.6208
## F-statistic: 40.29 on 4 and 92 DF,  p-value: < 2.2e-16
```

```
f15 <- update(f14, .~.-lbph)
summary(f15)
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol + lweight + svi, data = prostate)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.72964 -0.45764  0.02812  0.46403  1.57013
##
## Coefficients:
##             Estimate Std. Error t value       Pr(>|t|)
## (Intercept) -0.26809    0.54350  -0.493        0.62298
## lcavol       0.55164    0.07467   7.388 0.000000000063 ***
## lweight      0.50854    0.15017   3.386        0.00104 **
## svi          0.66616    0.20978   3.176        0.00203 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7168 on 93 degrees of freedom
## Multiple R-squared:  0.6264, Adjusted R-squared:  0.6144
## F-statistic: 51.99 on 3 and 93 DF,  p-value: < 2.2e-16
```

The model selected by the backward elimination:
lpsa = -0.268 + 0.552lcavol + 0.509lweight +0.666svi

Below result confirms the same variables removed by the backward elimination:

```
ols_step_backward_p(f1, prem=0.05)
```

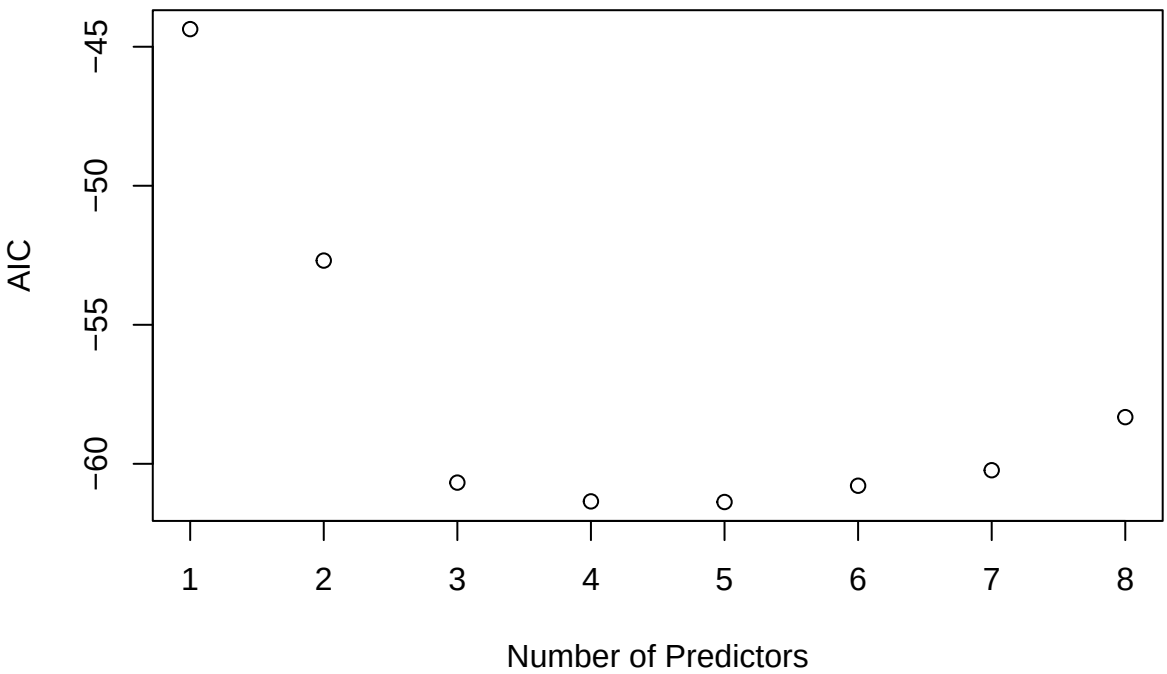
```
##
##
##                           Elimination Summary
## ---------------------------------------------------------------------------
##         Variable                   Adj.
## Step    Removed     R-Square    R-Square     C(p)       AIC        RMSE
## ---------------------------------------------------------------------------
##    1    gleason       0.6544      0.6273    7.0822    217.0428    0.7048
##    2    lcp           0.6493      0.6259    6.4020    216.4854    0.7061
##    3    pgg45         0.6441      0.6245    5.7150    215.8997    0.7073
##    4    age           0.6366      0.6208    5.6264    215.9223    0.7108
##    5    lbph          0.6264      0.6144    6.2169    216.5979    0.7168
## ---------------------------------------------------------------------------
```

b-) AIC

```
b <- regsubsets(lpsa ~ ., data=prostate)
rs <- summary(b)
rs$which
```

```
##   (Intercept) lcavol lweight   age  lbph   svi   lcp gleason pgg45
## 1        TRUE   TRUE   FALSE FALSE FALSE FALSE FALSE   FALSE FALSE
## 2        TRUE   TRUE    TRUE FALSE FALSE FALSE FALSE   FALSE FALSE
## 3        TRUE   TRUE    TRUE FALSE FALSE  TRUE FALSE   FALSE FALSE
## 4        TRUE   TRUE    TRUE FALSE  TRUE  TRUE FALSE   FALSE FALSE
## 5        TRUE   TRUE    TRUE  TRUE  TRUE  TRUE FALSE   FALSE FALSE
## 6        TRUE   TRUE    TRUE  TRUE  TRUE  TRUE FALSE   FALSE  TRUE
## 7        TRUE   TRUE    TRUE  TRUE  TRUE  TRUE  TRUE   FALSE  TRUE
## 8        TRUE   TRUE    TRUE  TRUE  TRUE  TRUE  TRUE    TRUE  TRUE
```

```
AIC<-97*log(rs$rss/97)+(2:9)*2
plot(AIC~I(1:8),ylab="AIC",xlab="Number of Predictors")
```




```
which.min(AIC)
```

```
## [1] 5
```

The plot above shows that AIC is minimized when there are 5 predictors, and the table suggests that we choose 'lcavol', 'lweight', 'age', 'lbph', and 'svi'.

Below stepwise method confirms the above result.

```
step(f1)
```

```
## Start:  AIC=-58.32
## lpsa ~ lcavol + lweight + age + lbph + svi + lcp + gleason +
##     pgg45
##
##           Df Sum of Sq    RSS     AIC
## - gleason  1    0.0412 44.204 -60.231
```

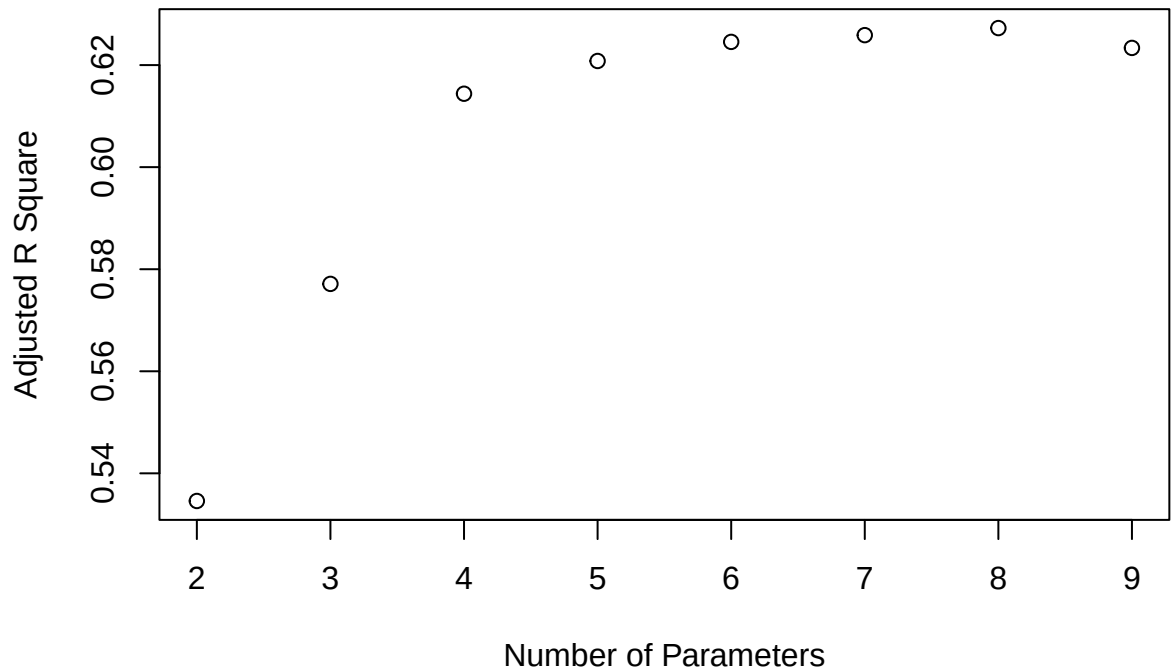
```
## - pgg45    1    0.5258 44.689 -59.174
## - lcp      1    0.6740 44.837 -58.853
## <none>                 44.163 -58.322
## - age      1    1.5503 45.713 -56.975
## - lbph     1    1.6835 45.847 -56.693
## - lweight  1    3.5861 47.749 -52.749
## - svi      1    4.9355 49.099 -50.046
## - lcavol   1   22.3721 66.535 -20.567
##
## Step:  AIC=-60.23
## lpsa ~ lcavol + lweight + age + lbph + svi + lcp + pgg45
##
##           Df Sum of Sq    RSS     AIC
## - lcp      1    0.6623 44.867 -60.789
## <none>                 44.204 -60.231
## - pgg45    1    1.1920 45.396 -59.650
## - age      1    1.5166 45.721 -58.959
## - lbph     1    1.7053 45.910 -58.560
## - lweight  1    3.5462 47.750 -54.746
## - svi      1    4.8984 49.103 -52.037
## - lcavol   1   23.5039 67.708 -20.872
##
## Step:  AIC=-60.79
## lpsa ~ lcavol + lweight + age + lbph + svi + pgg45
##
##           Df Sum of Sq    RSS     AIC
## - pgg45    1    0.6590 45.526 -61.374
## <none>                 44.867 -60.789
## - age      1    1.2649 46.131 -60.092
## - lbph     1    1.6465 46.513 -59.293
## - lweight  1    3.5647 48.431 -55.373
## - svi      1    4.2503 49.117 -54.009
## - lcavol   1   25.4189 70.285 -19.248
##
## Step:  AIC=-61.37
## lpsa ~ lcavol + lweight + age + lbph + svi
##
##           Df Sum of Sq    RSS     AIC
## <none>                 45.526 -61.374
## - age      1    0.9592 46.485 -61.352
## - lbph     1    1.8568 47.382 -59.497
## - lweight  1    3.2251 48.751 -56.735
## - svi      1    5.9517 51.477 -51.456
## - lcavol   1   28.7665 74.292 -15.871
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol + lweight + age + lbph + svi, data = prostate)
##
## Coefficients:
## (Intercept)       lcavol      lweight          age         lbph          svi
##     0.95100      0.56561      0.42369     -0.01489      0.11184      0.72095
```

lpsa = 0.951 + 0.566lcavol + 0.424lweight - 0.015age + 0.112lbph + 0.721svi

c-) Adjusted R_a^2

```
plot(2:9,rs$adjr2,xlab="Number of Parameters",ylab="Adjusted R Square")
```



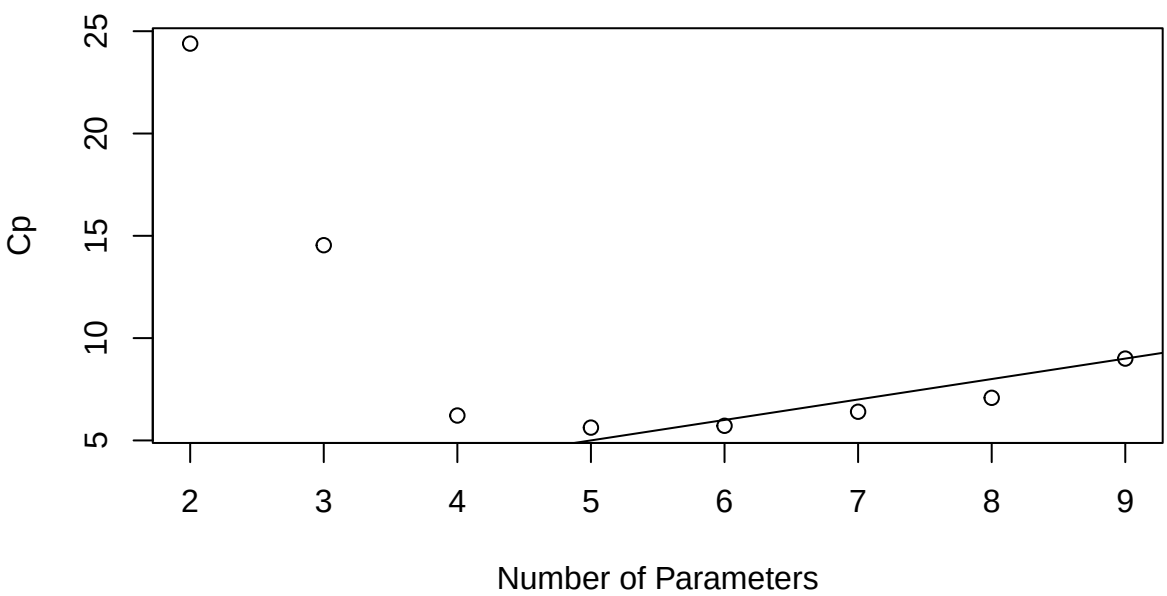

```
which.max(rs$adjr2)
```

```
## [1] 7
```

The adjusted R^2 is maximized when there are 8 parameters, but the marginal increase is very small once the number of parameters reaches 5 or 6.

d-) Mallows C_p

```
plot(2:9,rs$cp,xlab="Number of Parameters",ylab="Cp")
abline(0,1)
```



```r
ols_step_best_subset(f1)
```

```
##                    Best Subsets Regression
## ---------------------------------------------------------------
## Model Index    Predictors
## ---------------------------------------------------------------
##      1         lcavol
##      2         lcavol lweight
##      3         lcavol lweight svi
##      4         lcavol lweight lbph svi
##      5         lcavol lweight age lbph svi
##      6         lcavol lweight age lbph svi pgg45
##      7         lcavol lweight age lbph svi lcp pgg45
##      8         lcavol lweight age lbph svi lcp gleason pgg45
## ---------------------------------------------------------------
##
##                                                      Subsets Regression Summary
## ------------------------------------------------------------------------------------------------------------
##                         Adj.        Pred
## Model    R-Square    R-Square    R-Square     C(p)        AIC         SBIC        SBC        MSEP
## ------------------------------------------------------------------------------------------------------------
##   1        0.5394      0.5346      0.5179    24.3946    232.9080    -43.1213    240.6321    60.1553
##   2        0.5859      0.5771      0.5534    14.5415    224.5837    -51.1891    234.8825    54.6631
##   3        0.6264      0.6144      0.5869     6.2169    216.5979    -58.5271    229.4714    49.8518
##   4        0.6366      0.6208      0.5898     5.6264    215.9223    -58.8841    231.3705    49.0284
##   5        0.6441      0.6245      0.5882     5.7150    215.8997    -58.5526    233.9227    48.5502
##   6        0.6493      0.6259      0.5856     6.4020    216.4854    -57.6133    237.0831    48.3850
##   7        0.6544      0.6273      0.5835     7.0822    217.0428    -56.6319    240.2152    48.2125
##   8        0.6548      0.6234       0.576     9.0000    218.9522    -54.5019    244.6993    48.7211
## ------------------------------------------------------------------------------------------------------------
## AIC: Akaike Information Criteria
##  SBIC: Sawa's Bayesian Information Criteria
##  SBC: Schwarz Bayesian Criteria
##  MSEP: Estimated error of prediction, assuming multivariate normality
##  FPE: Final Prediction Error
##  HSP: Hocking's Sp
##  APC: Amemiya Prediction Criteria
```

We would choose between the six parameter, five predictor, model (Model 5) including 'lcavol', 'lweight', 'age', 'lbph', and 'svi', and the model also including 'pgg45' (Model 6). Model 7 is also below C_p=p line, but we prefer the ones with fewer predictors.

Model 5 is the closest to C_p=p line and has the smallest AIC, but we would still keep Model 6, which has similarly good numbers, as another candidate and take a deeper analysis (i.e. check assumptions) to select the final model.

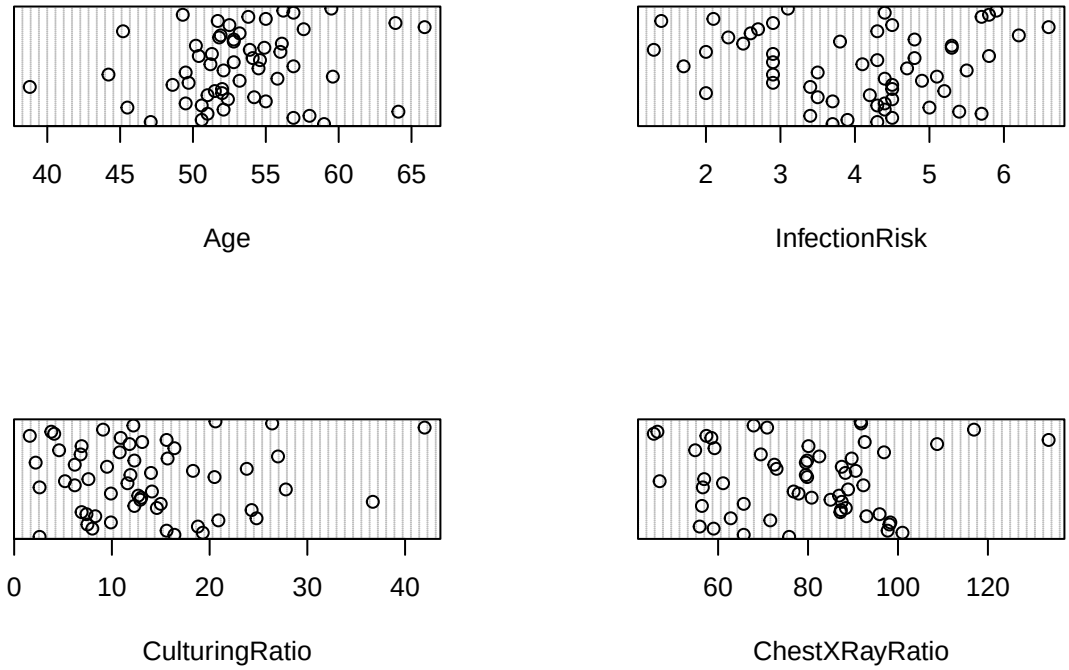
Problem 2

a-)

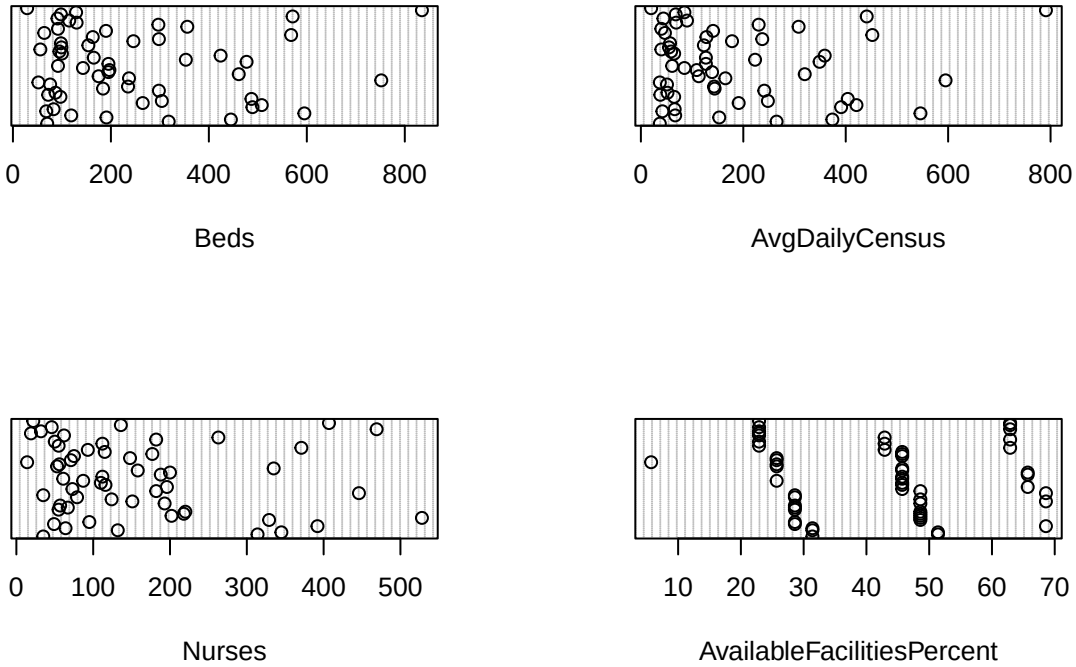
```r
SENIC <- getDataSet("SENIC.csv")
SENIC.r <- SENIC[,-c(1,8:9)] # drop 'MedicalSchoolAffiliation' and 'Region'

train.set <- SENIC.r[57:113,]
test.set <- SENIC.r[1:56,]
```

```r
par(mfrow=c(2,2))
dotchart(train.set$Age, xlab="Age")
dotchart(train.set$InfectionRisk, xlab="InfectionRisk")
dotchart(train.set$CulturingRatio, xlab="CulturingRatio")
dotchart(train.set$ChestXRayRatio, xlab="ChestXRayRatio")
```



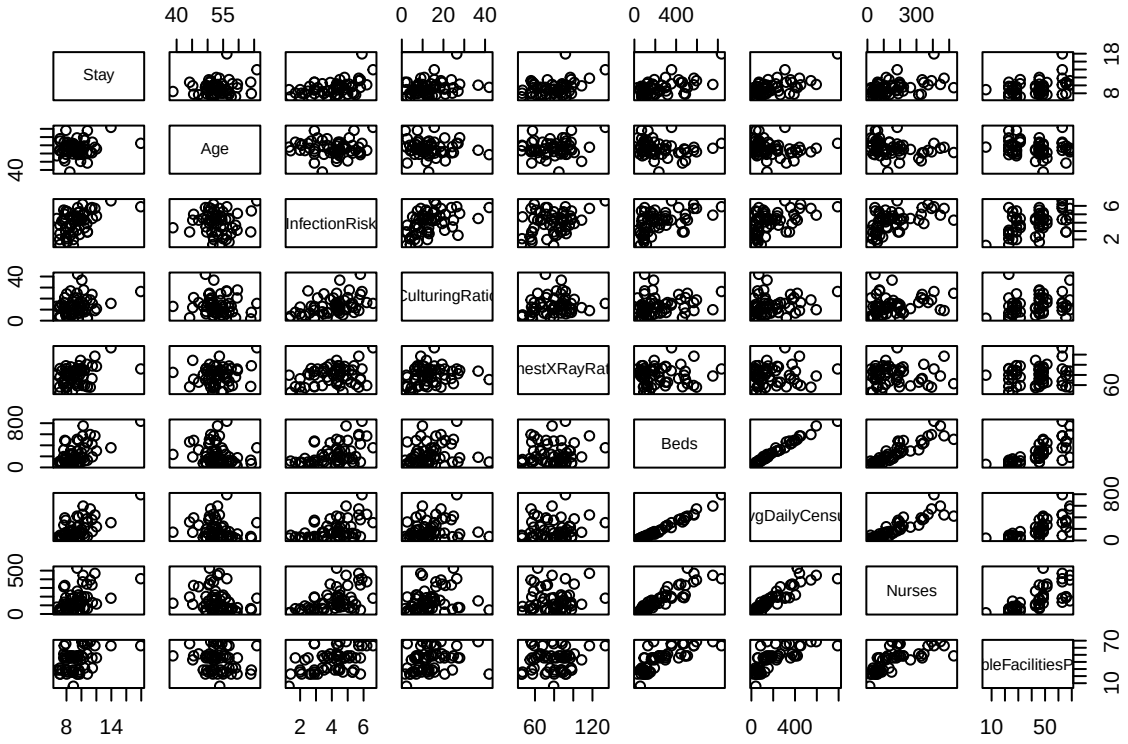

```r
dotchart(train.set$Beds, xlab="Beds")
dotchart(train.set$AvgDailyCensus, xlab="AvgDailyCensus")
dotchart(train.set$Nurses, xlab="Nurses")
dotchart(train.set$AvailableFacilitiesPercent, xlab="AvailableFacilitiesPercent")
```



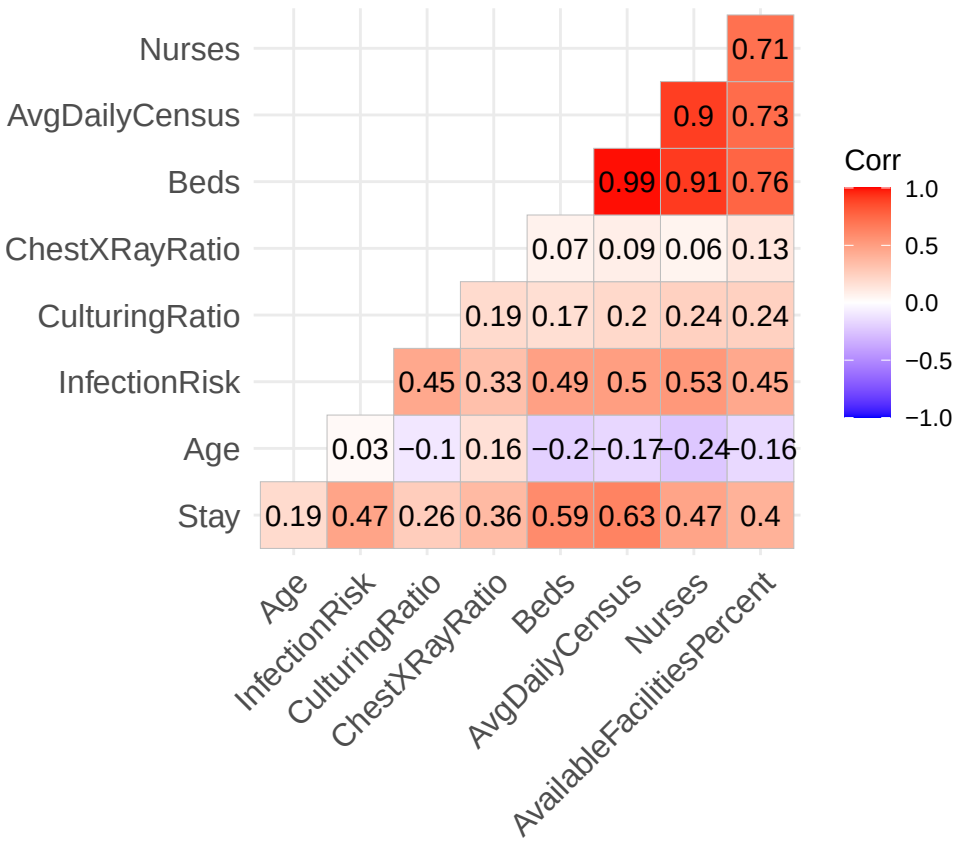
We see some outlying data points (e.g. one below 40 for Age, about 800 for Beds/AvgDailyCensus, below 10% for AvailableFacilitiesPercent), but there does not appear to be problematic features in the dot plots. It is notable that the variables like 'Beds', 'AvgDailyCensus', and 'Nurses' have data points concentrated on the left-hand side. 'AvailableFacilitiesPercent' almost looks like a categorical variable, but it is just due to the fact that there are many hospitals with the similar number of facilities and services.

b-)

```
plot(train.set)
```




```
corr <- round(cor(train.set),2)
ggcorrplot(corr, type="lower", lab=TRUE)
```



The following pairs of predictor variables have very strong linear pairwise associations:
Beds and AvgDailyCensus: r = 0.99
Beds and Nurses: r = 0.91
Nurses and AvgDailyCensus: r = 0.90

The following pairs also have quite strong linear pairwise associations:
AvailableFacilitiesPercent and Beds: r = 0.76
AvailableFacilitiesPercent and AvgDailyCensus: r = 0.73
AvailableFacilitiesPercent and Nurses: r = 0.71

c-)

```
f2 <- lm(log(Stay) ~ ., data=train.set)
ols_step_best_subset(f2)
```

```
##                                                Best Subsets Regression
## ---------------------------------------------------------------------------------------------------
## Model Index    Predictors
## ---------------------------------------------------------------------------------------------------
##      1         AvgDailyCensus
##      2         ChestXRayRatio AvgDailyCensus
##      3         Age ChestXRayRatio AvgDailyCensus
##      4         Age ChestXRayRatio AvgDailyCensus Nurses
##      5         Age CulturingRatio ChestXRayRatio AvgDailyCensus Nurses
##      6         Age InfectionRisk ChestXRayRatio Beds AvgDailyCensus Nurses
##      7         Age InfectionRisk CulturingRatio ChestXRayRatio AvgDailyCensus Nurses AvailableFacili
##      8         Age InfectionRisk CulturingRatio ChestXRayRatio Beds AvgDailyCensus Nurses AvailableF
## ---------------------------------------------------------------------------------------------------
##
```

```
##                                                    Subsets Regression Summary
## ---------------------------------------------------------------------------------------------------
##                          Adj.        Pred
## Model    R-Square    R-Square    R-Square     C(p)        AIC         SBIC         SBC        MSEP
## ---------------------------------------------------------------------------------------------------
##   1        0.3696      0.3582      0.2968    16.2329    -55.9935    -218.5217    -49.8643    1.1657
##   2        0.4676      0.4478      0.3849     7.4790    -63.6155    -225.5275    -55.4433    1.0032
##   3        0.5192      0.4919      0.4244     3.8112    -67.4264    -228.5634    -57.2111    0.9234
##   4        0.5369      0.5013      0.4324     3.8638    -67.5679    -228.1496    -55.3096    0.9068
##   5        0.5513      0.5073      0.4368     4.2839    -67.3665    -227.2958    -53.0652    0.8962
##   6        0.5576      0.5045      0.4033     5.5946    -66.1694    -225.5666    -49.8250    0.9017
##   7        0.5615      0.4988       0.407     7.1658    -64.6746    -223.5663    -46.2871    0.9123
##   8        0.5630      0.4901      0.3747     9.0000    -62.8712    -221.3255    -42.4406    0.9286
## ---------------------------------------------------------------------------------------------------
## AIC: Akaike Information Criteria
##  SBIC: Sawa's Bayesian Information Criteria
##  SBC: Schwarz Bayesian Criteria
##  MSEP: Estimated error of prediction, assuming multivariate normality
##  FPE: Final Prediction Error
##  HSP: Hocking's Sp
##  APC: Amemiya Prediction Criteria
```

```
Cp <- ols_step_best_subset(f2)[3:5,7]
p <- c(4,5,6)
cpp <- data.frame(cbind(Cp, p))
rownames(cpp) <- c("Model 3","Model 4","Model 5")
kable(cpp, digits=3, align="c", caption="Cp and p Comparison", booktabs=TRUE) %>%
  kable_styling(latex_options="HOLD_position")
```

Table 1: Cp and p Comparison

	Cp	p
Model 3	3.811	4
Model 4	3.864	5
Model 5	4.284	6

According to the table above, Model 3, 4, and 5 are the three best subsets for the C_p criterion. Among these three subsets, Model 3 appears to have the smallest bias since Model 3 has C_p=3.8112 and p=4. The gap is only 0.1888, and Model 4 and 5 have bigger gaps between C_p and p. Also, we want C_p to be small and to be close to p.

d-)

```
f2.train <- lm(log(Stay) ~ Age+ChestXRayRatio+AvgDailyCensus, data=train.set)
s1 <- summary(f2.train)
s1
```

```
##
## Call:
## lm(formula = log(Stay) ~ Age + ChestXRayRatio + AvgDailyCensus,
##     data = train.set)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -0.25937 -0.08657  0.02955  0.07747  0.21522
##
## Coefficients:
##                 Estimate Std. Error t value      Pr(>|t|)
## (Intercept)    1.4055759  0.2045023   6.873 0.00000000722 ***
## Age            0.0089343  0.0037461   2.385       0.02069 *
## ChestXRayRatio 0.0027050  0.0009643   2.805       0.00702 **
## AvgDailyCensus 0.0006738  0.0001049   6.420 0.00000003859 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1272 on 53 degrees of freedom
## Multiple R-squared:  0.5192, Adjusted R-squared:  0.4919
## F-statistic: 19.07 on 3 and 53 DF,  p-value: 0.00000001614
```

```
f2.test <- lm(log(Stay) ~ Age+ChestXRayRatio+AvgDailyCensus, data=test.set)
s2 <- summary(f2.test)
s2
```

```
##
## Call:
## lm(formula = log(Stay) ~ Age + ChestXRayRatio + AvgDailyCensus,
##     data = test.set)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.31188 -0.10566 -0.00886  0.09251  0.50058
##
## Coefficients:
##                 Estimate Std. Error t value   Pr(>|t|)
## (Intercept)    1.4249909  0.2872850   4.960 0.00000792 ***
## Age            0.0091956  0.0048558   1.894    0.06383 .
## ChestXRayRatio 0.0035052  0.0010068   3.482    0.00102 **
## AvgDailyCensus 0.0003610  0.0001431   2.522    0.01476 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1497 on 52 degrees of freedom
## Multiple R-squared:  0.2934, Adjusted R-squared:  0.2526
## F-statistic: 7.196 on 3 and 52 DF,  p-value: 0.0003955
```

```
b1.co <- c(s1$coefficients[2,1], s2$coefficients[2,1])
b2.co <- c(s1$coefficients[3,1], s2$coefficients[3,1])
b3.co <- c(s1$coefficients[4,1], s2$coefficients[4,1])

b1.sd <- c(s1$coefficients[2,2], s2$coefficients[2,2])
b2.sd <- c(s1$coefficients[3,2], s2$coefficients[3,2])
b3.sd <- c(s1$coefficients[4,2], s2$coefficients[4,2])

tbl1 <- data.frame(rbind(b1.co, b2.co, b3.co))
rownames(tbl1) <- c("b1","b2","b3")
colnames(tbl1) <- c("Train","Test")

tbl2 <- data.frame(rbind(b1.sd, b2.sd, b3.sd))
rownames(tbl2) <- c("b1","b2","b3")
```

```
colnames(tbl2) <- c("Train","Test")

kable(tbl1, digits=5, align="c", caption="Regression Coefficients", booktabs=TRUE) %>%
  kable_styling(latex_options="HOLD_position")
```

Table 2: Regression Coefficients

	Train	Test
b1	0.00893	0.00920
b2	0.00271	0.00351
b3	0.00067	0.00036

```
kable(tbl2, digits=5, align="c", caption="Standard Deviations of Regression Coefficients", booktabs=TRU
  kable_styling(latex_options="HOLD_position")
```

Table 3: Standard Deviations of Regression Coefficients

	Train	Test
b1	0.00375	0.00486
b2	0.00096	0.00101
b3	0.00010	0.00014

The estimated regression coefficients and their standard deviations do not differ much between the two models. But, for the validation set, the summary shows that the p-value for 'Age' is bigger than 0.05, making it insignificant for the regression model using the validation set whereas it is significant for that using the training set.

e-)

```
MSE.train <- anova(f2.train)$`Mean Sq`[4]
MSE.test <- anova(f2.test)$`Mean Sq`[4]
tbl3 <- data.frame(MSE.train, MSE.test)

tbl4 <- data.frame(s1$r.squared, s2$r.squared)
colnames(tbl4) <- c("Rsquared(Train)","Rsquared(Test)")

kable(tbl3, digits=5, align="c", caption="MSE", booktabs=TRUE) %>%
  kable_styling(latex_options="HOLD_position")
```

Table 4: MSE

MSE.train	MSE.test
0.01619	0.02241

```
kable(tbl4, digits=5, align="c", caption="Coefficients of Multiple Determination", booktabs=TRUE) %>%
  kable_styling(latex_options="HOLD_position")
```

Table 5: Coefficients of Multiple Determination

Rsquared(Train)	Rsquared(Test)
0.51916	0.29337

MSEs do not differ much, but R^2 is significantly reduced for the validation set compared to the training set. Both models are significant, but considering very different R^2 and bigger p-value of 'Age' for the model fitted to the validation set, we need a further analysis. It could be the case that some other subset model fits better to the validation set.

Problem 3

```
data(teengamb)
f3 <- lm(gamble ~ ., data=teengamb)
summary(f3)
```

```
##
## Call:
## lm(formula = gamble ~ ., data = teengamb)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -51.082 -11.320  -1.451   9.452  94.252
##
## Coefficients:
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept)  22.55565   17.19680   1.312    0.1968
## sex         -22.11833    8.21111  -2.694    0.0101 *
## status        0.05223    0.28111   0.186    0.8535
## income        4.96198    1.02539   4.839 0.0000179 ***
## verbal       -2.95949    2.17215  -1.362    0.1803
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.69 on 42 degrees of freedom
## Multiple R-squared:  0.5267, Adjusted R-squared:  0.4816
## F-statistic: 11.69 on 4 and 42 DF,  p-value: 0.000001815
```

```
stepwise.result <- ols_step_both_p(f3, pent=0.1, prem=0.05)
stepwise.result
```

```
##
##                              Stepwise Selection Summary
## -----------------------------------------------------------------------------------------
##                      Added/                    Adj.
## Step    Variable    Removed     R-Square    R-Square     C(p)        AIC        RMSE
## -----------------------------------------------------------------------------------------
##    1     income     addition       0.387       0.373    11.4010    439.7158    24.9482
##    2      sex       addition       0.501       0.479     3.2480    432.0070    22.7543
## -----------------------------------------------------------------------------------------
```

```
summary(stepwise.result$model)
```

```
##
## Call:
## lm(formula = paste(response, "~", paste(preds, collapse = " + ")),
##     data = l)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -49.757 -11.649   0.844   8.659 100.243
##
## Coefficients:
##             Estimate Std. Error t value   Pr(>|t|)
## (Intercept)    4.041      6.394   0.632    0.53070
## income         5.172      0.951   5.438 0.00000224 ***
## sex          -21.634      6.809  -3.177    0.00272 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.75 on 44 degrees of freedom
## Multiple R-squared:  0.5014, Adjusted R-squared:  0.4787
## F-statistic: 22.12 on 2 and 44 DF,  p-value: 0.0000002243
```

gamble = 4.041 + 5.172income - 21.634sex

We see below that adding 'verbal' or 'status' does not make either variable significant, so the above final model with only 'income' and 'sex' is confirmed as the final one, and both variables are significant.

```r
f3.2 <- lm(gamble ~ income+sex+verbal, data=teengamb)
summary(f3.2)
```

```
##
## Call:
## lm(formula = gamble ~ income + sex + verbal, data = teengamb)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.639 -11.765  -1.594   9.305  93.867
##
## Coefficients:
##             Estimate Std. Error t value   Pr(>|t|)
## (Intercept)  24.1390    14.7686   1.634     0.1095
## income        4.8981     0.9551   5.128 0.00000664 ***
## sex         -22.9602     6.7706  -3.391     0.0015 **
## verbal       -2.7468     1.8253  -1.505     0.1397
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.43 on 43 degrees of freedom
## Multiple R-squared:  0.5263, Adjusted R-squared:  0.4933
## F-statistic: 15.93 on 3 and 43 DF,  p-value: 0.0000004148
```

```r
f3.3 <- lm(gamble ~ income+sex+status, data=teengamb)
summary(f3.3)
```

```
##
## Call:
## lm(formula = gamble ~ income + sex + status, data = teengamb)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -48.682 -12.169  -0.268   9.161  97.728
##
## Coefficients:
##             Estimate Std. Error t value  Pr(>|t|)
## (Intercept)  13.0315    15.8676   0.821   0.41603
## income        4.9280     1.0352   4.760 0.0000221 ***
## sex         -24.3393     8.1274  -2.995   0.00454 **
## status       -0.1496     0.2413  -0.620   0.53856
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.92 on 43 degrees of freedom
## Multiple R-squared:  0.5058, Adjusted R-squared:  0.4713
## F-statistic: 14.67 on 3 and 43 DF,  p-value: 0.000001014
```